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How robust are the parameter constraints extending the LCDM model?

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ABSTRACT

Comprehensive overview of cosmological parameters from current CMB (Planck, ACT, SPT) + CMB lensing (4-point) + DESI BAO. Discusses correlations: N_eff vs Omega_b h2, N_eff vs n_s, H0 vs N_eff, r_drag vs N_eff. Provides marginalized distance priors for model testing.

What was actually measured (numbers):

Reports 68% and 95% CL for 6 base LCDM + extended, correlation matrices, r_drag sensitivity to N_eff

Devices / Protocol:

Planck PR4, ACT DR6, SPT-3G, CMB lensing, DESI DR2 BAO (9 bins 0.295 to 2.33)

Replication across teams / Independence:

Multi-dataset combination shows low-z measurements break degeneracies

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